Chapter Four

Internet of Things (IOT)

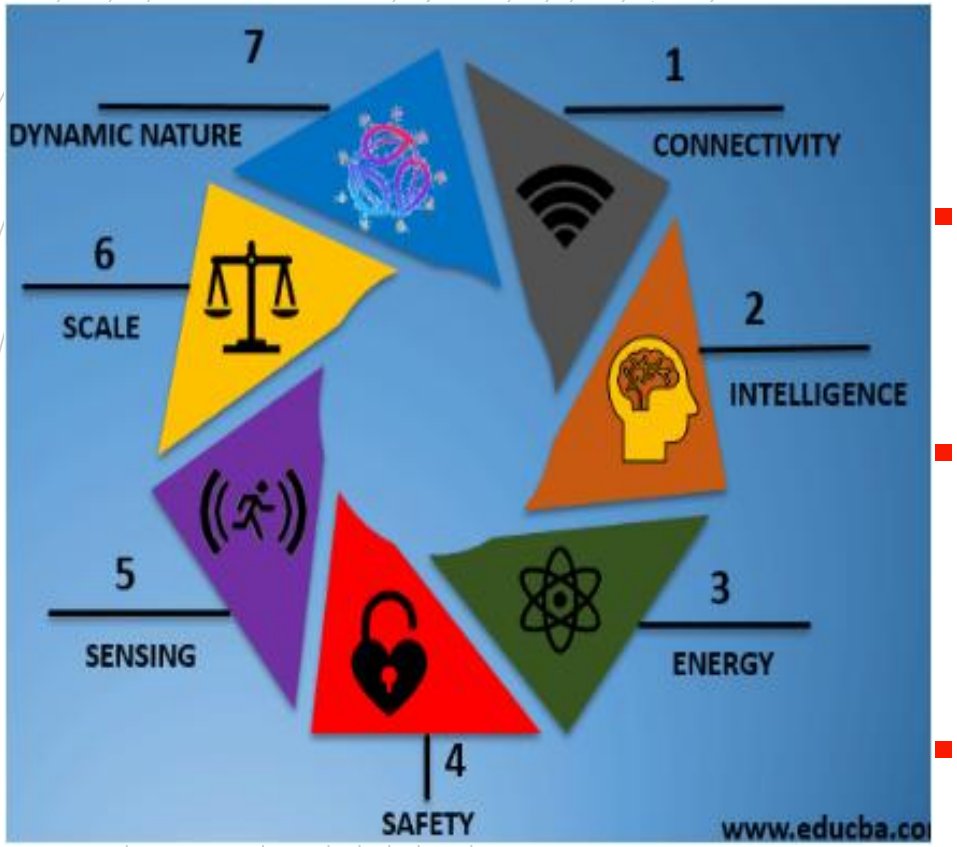
Outline

- What is IoT?
- Important features of IoT
- History of IoT
- Advantages and disadvantages of IoT
- Challenges of IoT
- How IoT works?
- Architecture of IoT
- Device and networks
- Tools and platforms of IoT
- Applications of IoT in homes, cities and farming

What is IoT?

- IoT is the networking **of smart objects**, meaning a huge number of devices intelligently communicating; Internet Architecture Board's (IAB)
- IoT is a framework of all things that have a representation in the presence of the internet and enables Machine-to-Machine (M2M) communication.
- IoT according to the 2020 conceptual framework is expressed through a simple formula such as:
  - **IoT= Services+ Data+ Networks + Sensors**
- Generally, IoT is the network of physical objects or "things" embedded with electronics, software, sensors, and network connectivity, which enables these objects to collect and exchange data.

The most **important features** of IoT



- **Artificial intelligence**:
  - IoT essentially makes virtually anything "smart"
  - It enhances life with the power of data collection, AI algorithms, and networks
- **Connectivity:**
  - Networks can exist on a much smaller and cheaper scale
  - IoT creates these small networks b/n its system devices
- **Sensors**:
  - Defining instruments these are capable of understanding real-world and integration with network
  - IoT loses its distinction without sensors
- **Active engagement:**
  - IoT introduces a new paradigm for active content, product, or service engagement
- **Small device:**
  - IoT exploits purpose-built small devices to deliver its precision, scalability, and versatility

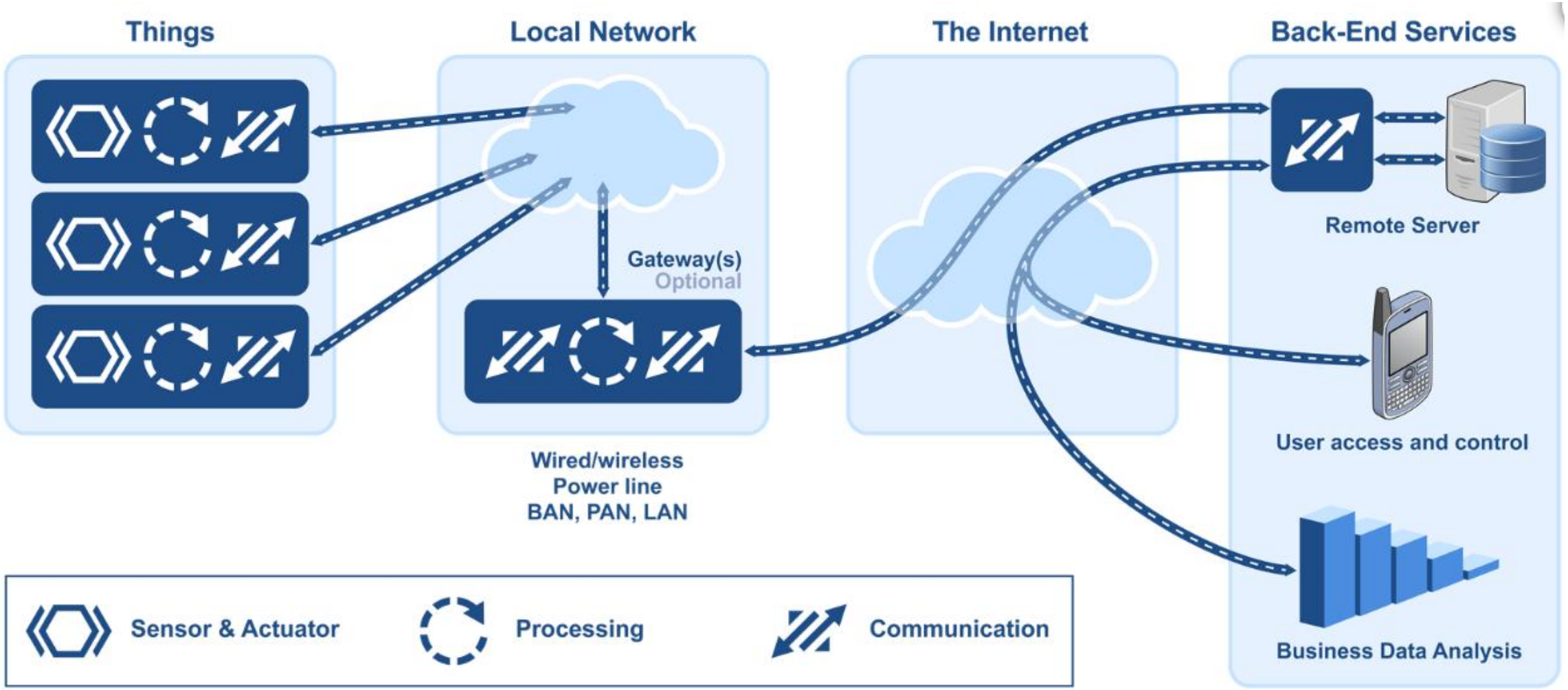
Components of IoT

IoT= Services+ Data+ Networks + Sensors(Things)

- IoT, refers to the set of devices and systems that interconnect real-world sensors and actuators to the Internet.
- **Things**: natural or man-made object that can be assigned an IP address and is able to transfer data over a network
  - ✓ **person** with a **heart monitor implant**,
  - ✓ a **farm animal** with a **biochip transponder**,
  - ✓ an **automobile** that has **built-in sensor**
- **Local network**
  - ✓ Gateway which translates proprietary communication protocols to Internet Protocol
- **Internet** - the Global Network
- **Back-end services**
  - ✓ Enterprise data systems, or PCs and mobile devices.
- **Data** – information captured by sensors

Components of IoT

IoT= Services+ Data+ Networks + Sensors(Things)



Internet of Things (IOT)

History of IoT

History of IoT

- Telegraph (the first landline) - 1830s
- "wireless telegraphy," the first radio voice transmission - June 3, 1900
- Computers development - 1950s.
- The Internet, by DARPA (Defense Advanced Research Projects Agency) in 1962 and evolved into ARPANET in 1969 – commercialized in 1980s.
- Global Positioning Satellites (GPS) - 1993, by the Department of Defense
- The first examples of an Internet of Things is in the early 1980s and was a Coca Cola machine, located at the Carnegie Melon University .
- **Kevin Ashton**, the Executive Director of Auto-ID Labs at MIT, was the first to describe the Internet of Things, during his 1999 speech.
- 2013, the Internet of Things had evolved into a system using multiple technologies

History of Internet of Things (IOT)

- IoT is a network of devices that can sense, accumulate and transfer data over the internet without any human intervention
- Its new and advanced elements bring major changes in the delivery of products, goods, and services; and the social, economic, and political impact of those changes

Pre-Internet

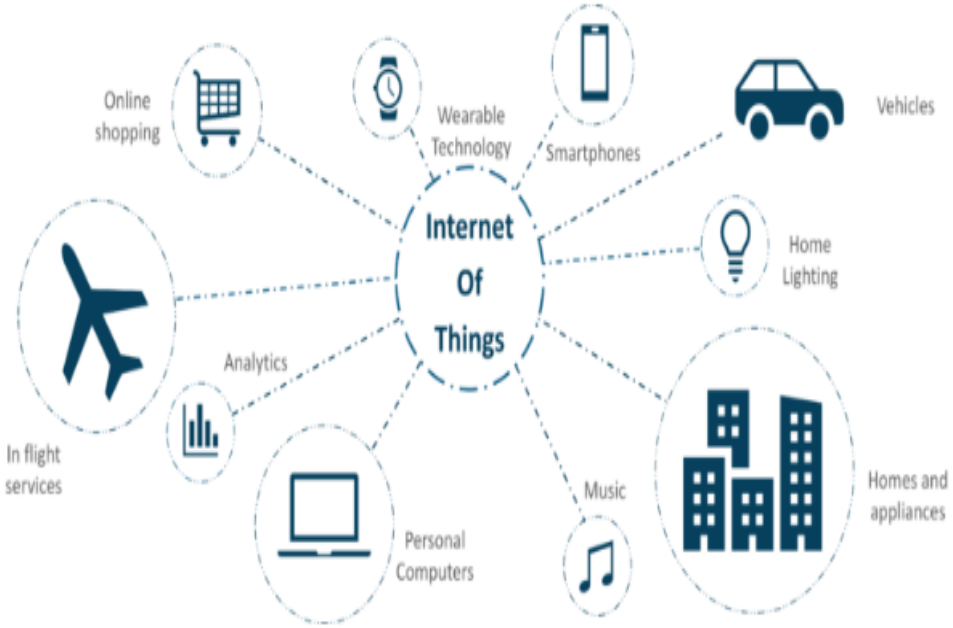
Dawn of Internet

Internet of People

Internet of Things

Evolution of the Internet

- IoT evolved to **machine-to-machine (M2M)** communication, i.e., machines connecting to each other via a network without human interaction.
- M2M refers to connecting a device to the cloud, managing it and collecting data
- In 2013, the IoT had evolved into a system using multiple technologies, includes
  - Internet to wireless communication
  - micro-electromechanical systems (MEMS)
  - The traditional fields of automation
  - Wireless Sensor Networks (WSN),
  - GPS,
  - Control systems like SCADA (Supervisory Control and Data Acquisition)
- The first to describe the Internet of Things, during his 1999 speech by **Kevin Ashton**, at MIT
  - He said Radio Frequency Identification (**RFID)** was a prerequisite
  - He concluded if all devices were "**tagged,"** computers could manage, track, and inventory
  - Tagging of things ➔ by digital watermarking, barcodes, and QR codes



IoT – Advantages

- **Improved Customer Engagement**: remove blind-spots and passive engagement
- **Technology Optimization** : D2D
- **Reduced Waste** : IoT provides real-world information leading to the more effective management of resources
- **Enhanced Data Collection**: allows an accurate picture of everything

IoT – Disadvantages

- **Privacy and security** : Potential hacker may increase
- If **bug** in the system, every connected device will **corrupt**
- **Absence of standards** : affects device interoperability
- Big Data Management
- Device **Energy** Consumption

Internet of Things (IOT)

Overview of IoT

Challenges of IoT

- **Security:** exposed to various kinds of attackers
- **Privacy:** provides substantial personal data in extreme detail
- **Complexity:** design, deployment, and maintenance
- **Flexibility:** concerns on how to integrate easily, lack of standards
- **Compliance:** complexity makes it challenging to comply with regulation

How IoT works?

How does IoT work?

An IoT ecosystem consists:

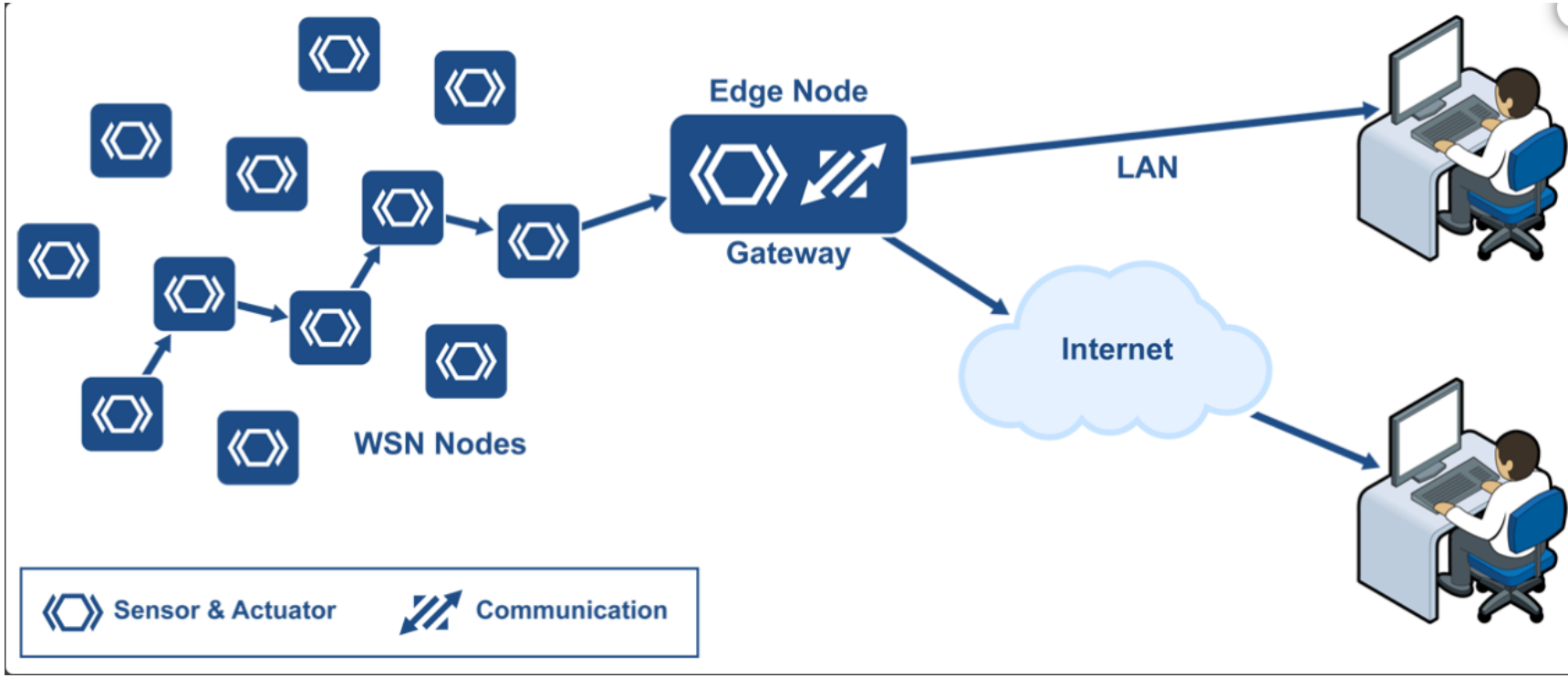
- Web-enabled smart devices that use embedded processors and connected to sensors, actuators and internet
- Sensors to collect data from the environment and send it back to the end smart device for analysis.
- Communication protocols used by the smart device to send data to the cloud to be analyze and provide remote access

The devices do most of the work without human intervention, although people can interact with the devices.

- ✓ to set up devices
- ✓ to give special instructions
- ✓ access data

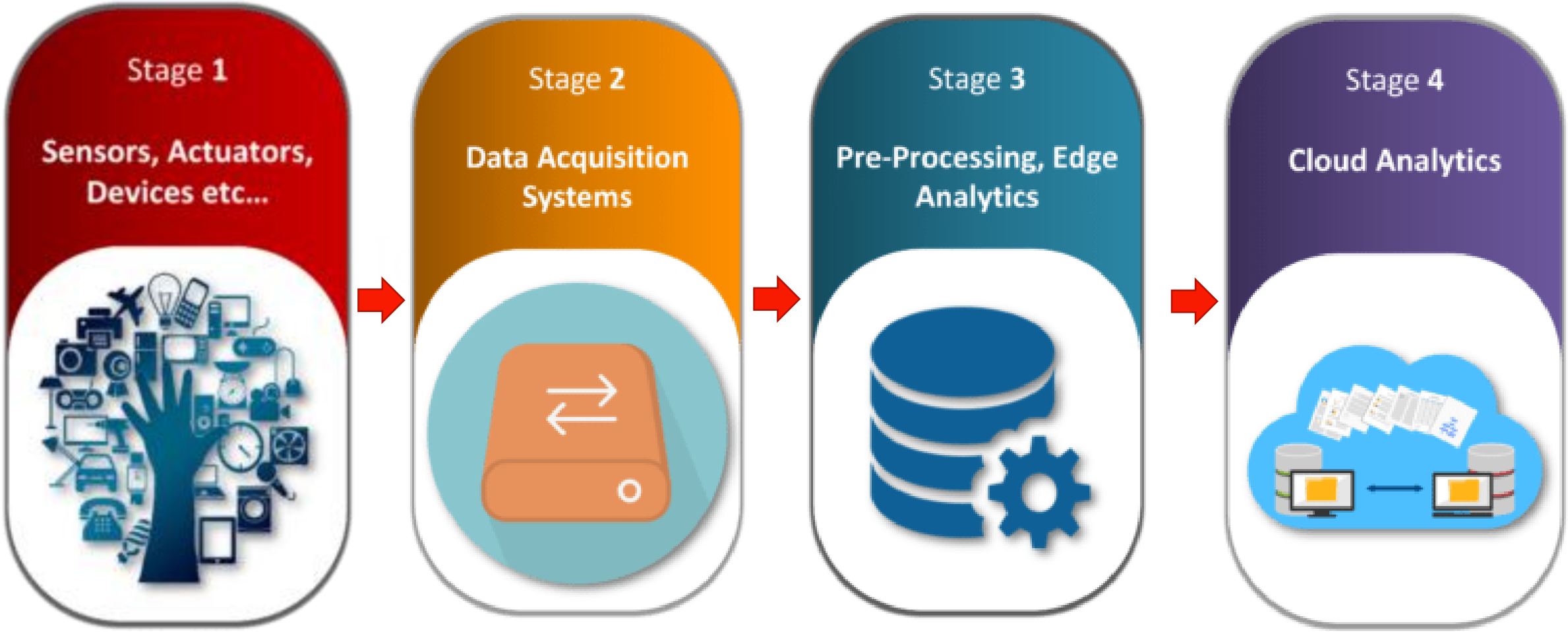
How IoT works…Cont'… Example

- Smart devices with embedded processors, sensors & communication hardware collect, send and act on data they acquire from their environments.
- Smart devices share collected data through IoT **gateway** or another **edge device** where data is either sent to the cloud to be **analyzed** or **analyzed locally**.



Wireless sensor network installed in a factory, connected to the Internet via a gateway

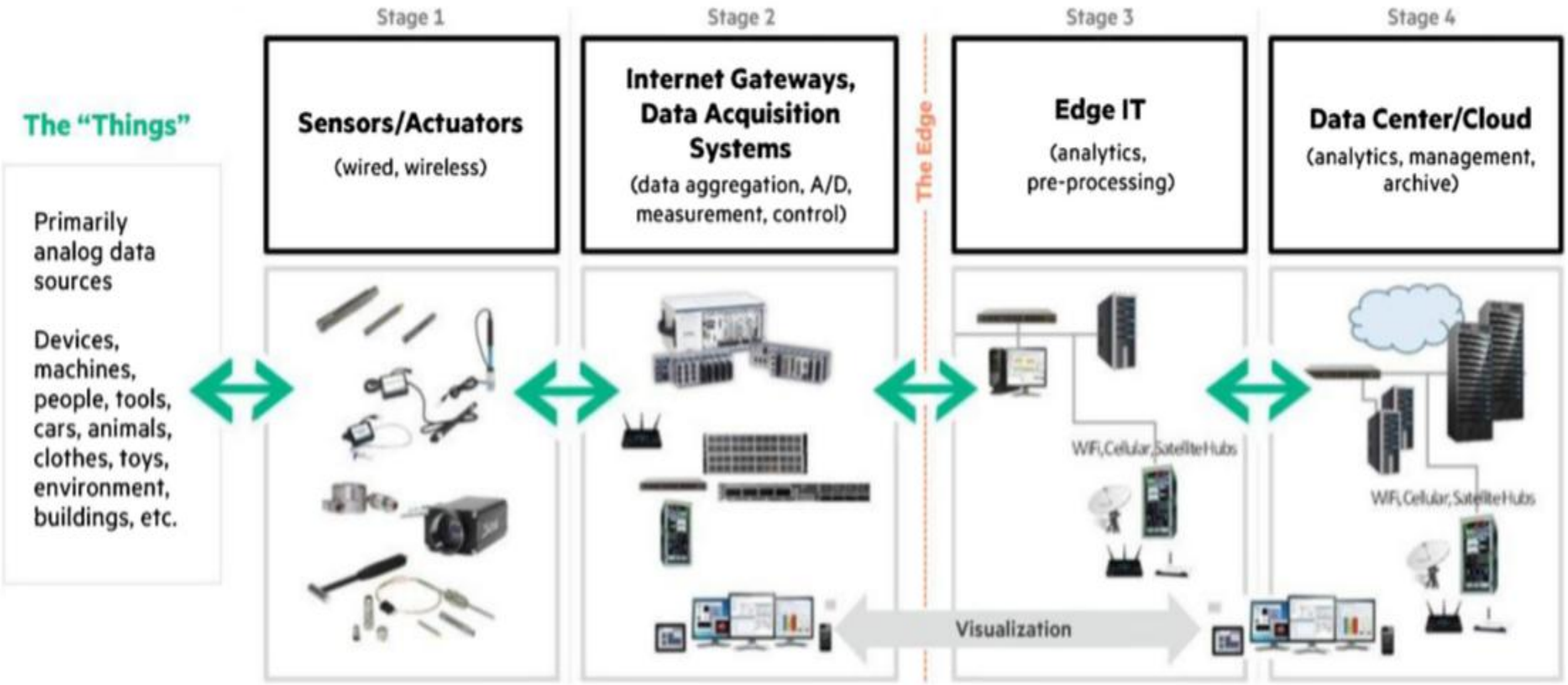
How IoT works…the 4 stages



How IoT works…the 4 stages

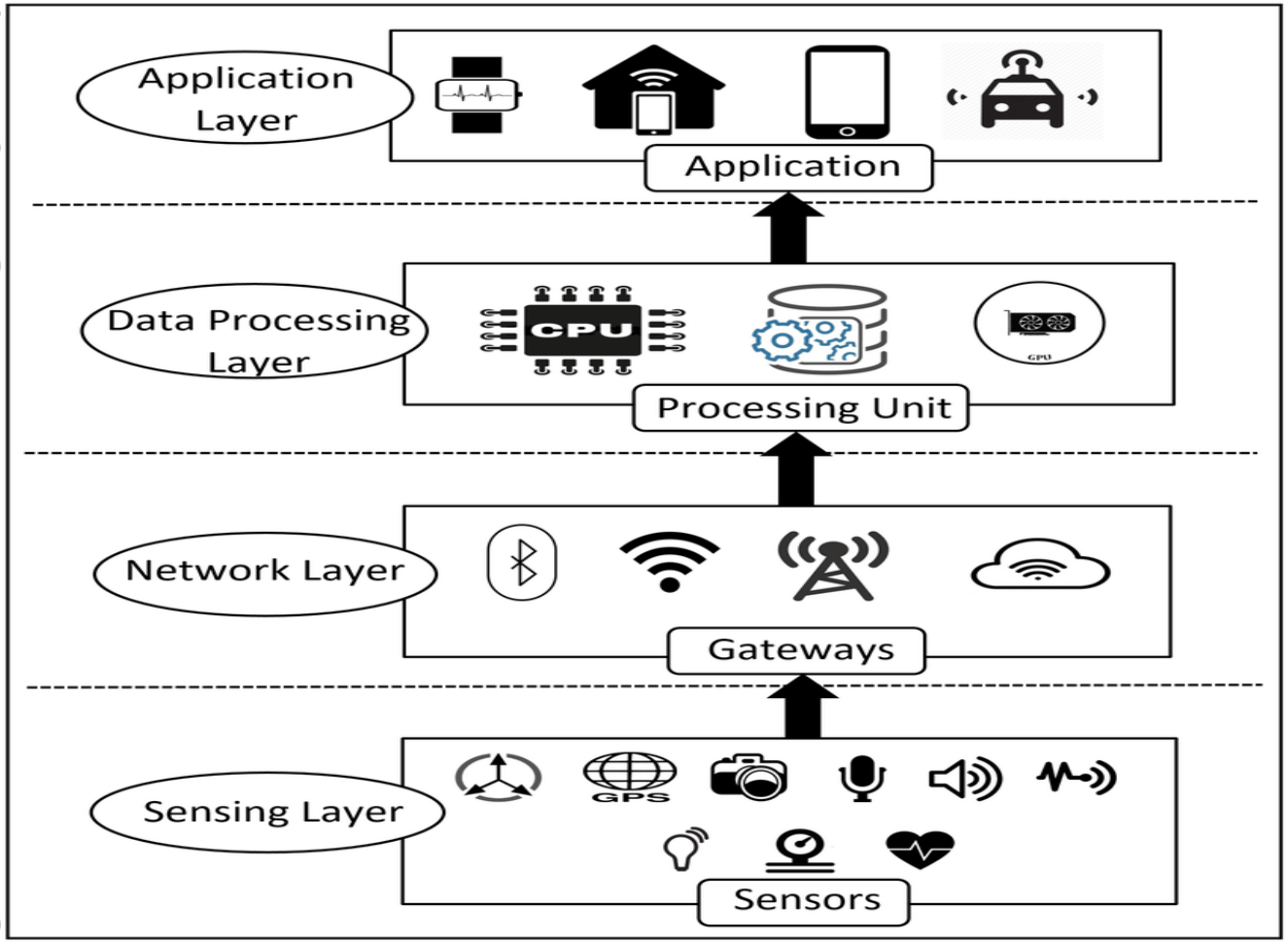
- **Stage 1 (Sensors/Actuators)**
  - A thing in the context of "Internet of Things", should be equipped with sensors and actuators thus giving the ability to emit, accept and process signals
- **Stage 2 (Data Acquisition Systems):**
  - The data from the sensors starts in analogue form which needs to be aggregated and converted into digital streams for further processing.
  - This stage performs these data aggregation and conversion functions
- **Stage 3 (Edge Analytics)**
  - Once IoT data has been digitized and aggregated, it may require further processing before it enters the data center, this is where Edge Analytics comes in
- **Stage 4 (Cloud Analytics)**
  - Data that needs more in-depth processing gets forwarded to physical data centers or cloud-based systems.

How IoT works…Cont'…



Architecture of IoT

- The architecture of IoT devices comprises four major components: sensing, network, data processing, and application layers



An IoT device consists:

- Hardware,
- Software,
- Network connectivity, and
- Sensors.

Hence, the architecture of IoT comprises:

- Sensing layer,
- Network layer,
- Data processing layer, and
- Application layers

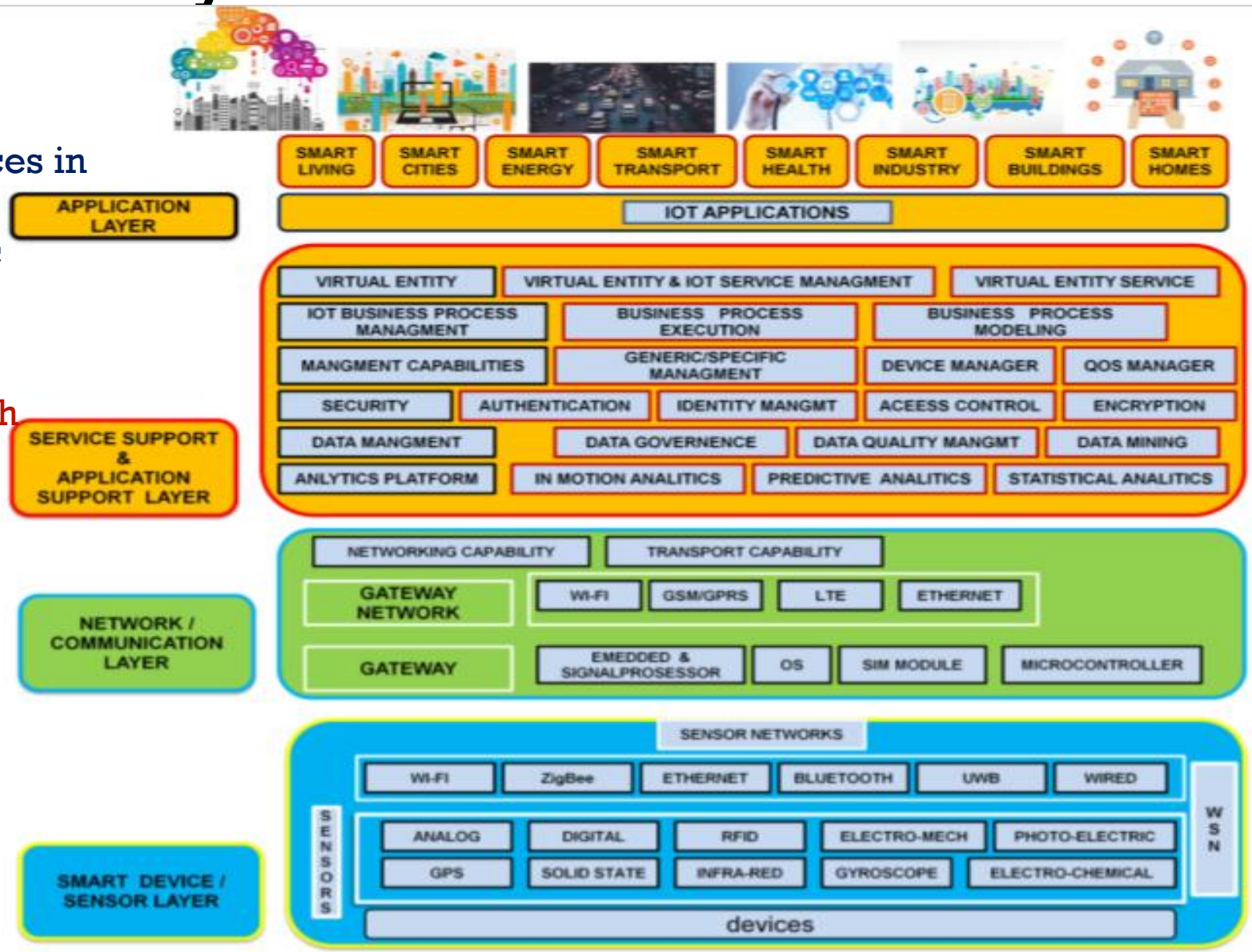
IoT Layered Architecture

"smart" environments/spaces in building, city, Health care, agriculture, etc

renders the processing of information possible through Analytics, Security controls, Process modeling and Management of devices

transport medium for the data generated by sensor

made up of smart objects integrated with sensors,



Internet of Things (IOT)

Architecture

Sensing Layer

- Used to identify any phenomena in the devices' peripheral and obtain data from the real world
- This layer consists of several sensors integrated through **sensor hubs** (point multiple sensors connected)
- ***Actuators :*** change the physical conditions that generate the data
  - eg. shut off a power supply, adjust an airflow valve
- Sensors in IoT devices can be classified into three broad categories

  A. Motion Sensors: *sense a change in motion and orientation*

  ✓ *linear* and *angular* motions

  B. Environmental Sensors : *sense a change in environmental parameters*

  ✓ Light sensors, Pressure sensors

  C. Position sensors : *sense physical position & location*

  ✓ GPS sensors ➔ navigation

  ✓ Magnetic sensors (digital Compass) ➔ fix orientation

Internet of Things (IOT)

Architecture

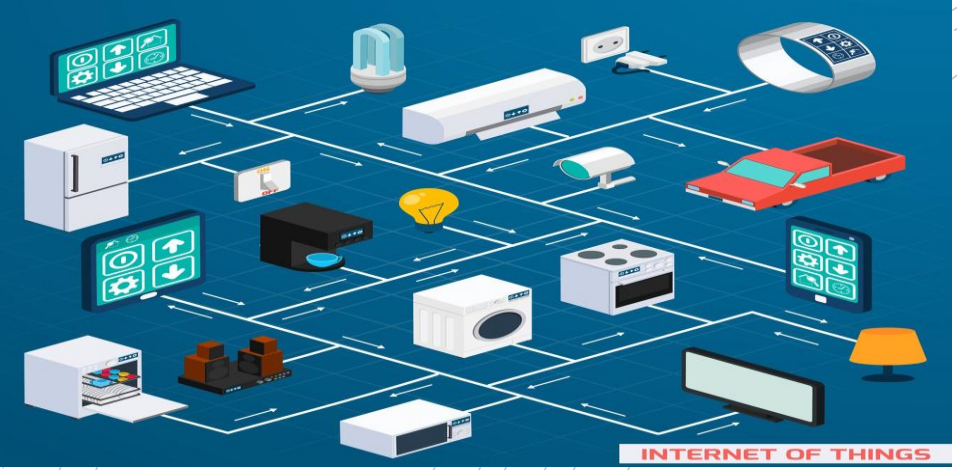
- **Network Layer**
- Acts as a communication channel to transfer data, collected in the sensing layer, to other connected devices
- The network layer is implemented by using diverse communication technologies (e.g., Wi-Fi, Bluetooth, Zigbee, Z-Wave, LoRa, cellular network, etc.)
- **Data Processing Layer**
- Takes data collected in the sensing layer and **analyses t**he data to make decisions based on the result
- The data processing layer also saves the result of the previous analysis and may share the result with other connected devices
- **Application Layer**
- Implements and presents the results of the data processing layer to accomplish **the intended applications** of IoT devices
- User-centric layer that executes various tasks for the users
- Smart transportation, smart home, personal care, healthcare…

Internet of Things (IOT)

Devices and Networks

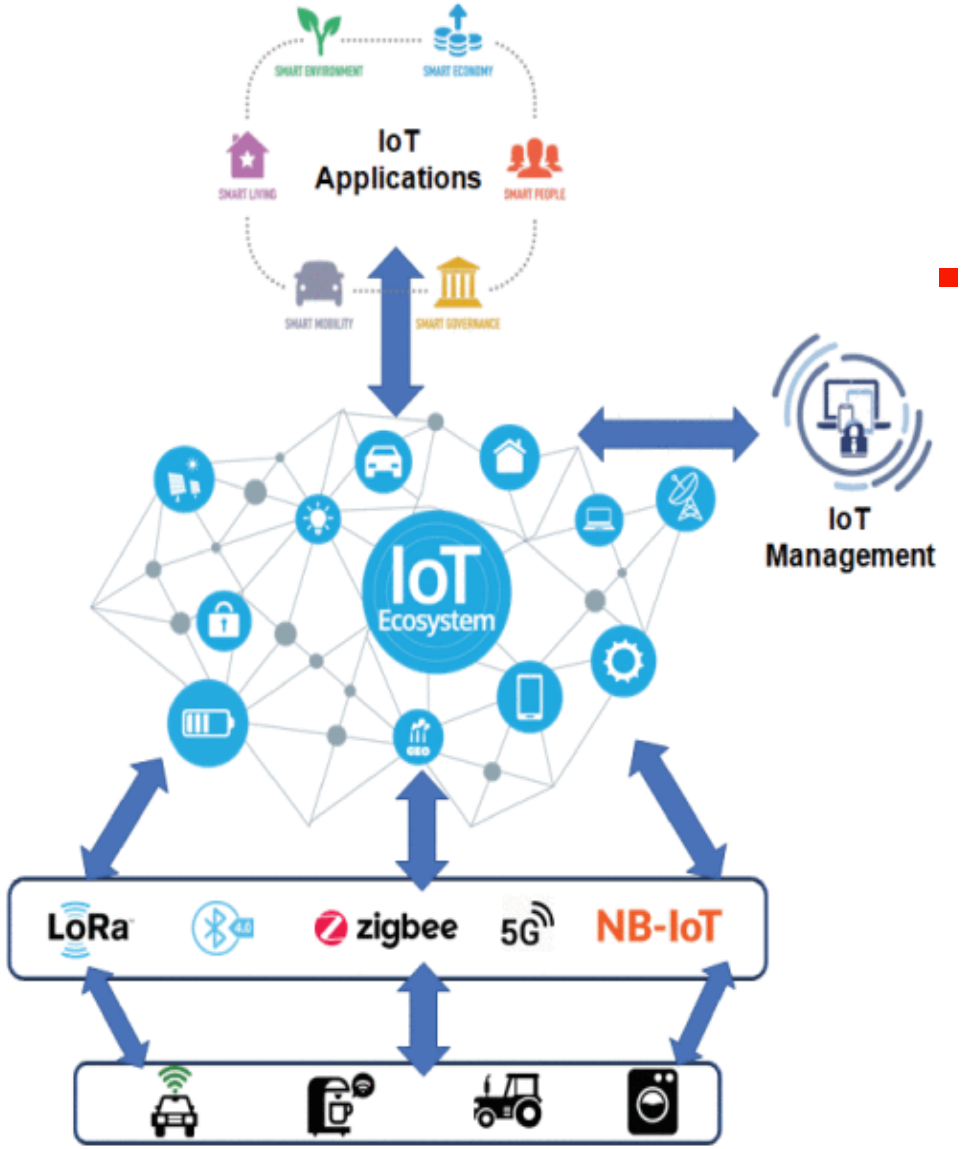
Devices and Networks

- IoT devices can be categorized into three main groups: consumer, enterprise and industrial
- **Consumer** connected devices include smart TVs, smart speakers, toys, wearables, and smart appliances
- Smart meters, commercial security systems and smart city technologies ➔ used to monitor traffic and weather conditions are examples of **industrial** IoT
- In **enterprise**, smart air conditioning, smart thermostats, smart lighting, and smart security, span home, smart sensors located in a conference room are used

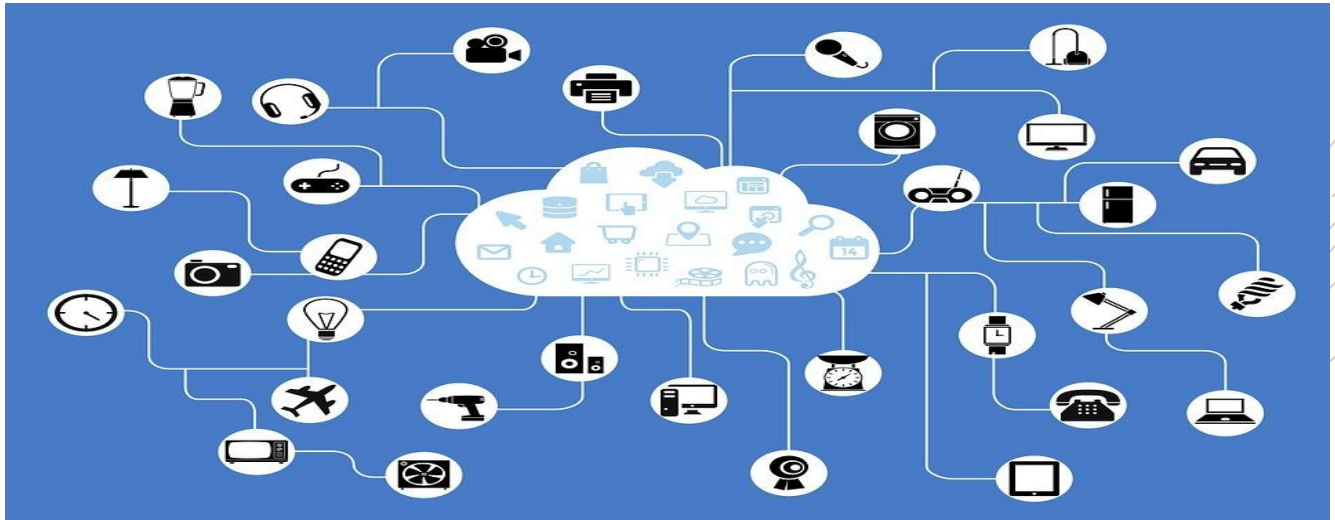


Devices and Networks

- IoT network devices use constrained resources → power, processing speed, memory
- Those devices may be deployed
  - Over large areas like smart cities, industrial plants,
  - Hard-to-reach areas like war, hazardous place
- Efficient management of IoT networks requires
  - Low power IoT devices
  - Deployment complexity
- Developing an IoT network management solution is not an easy task because of the
  - increasing number of connected devices characterized by their heterogeneity
  - intrinsic constraints of IoT networks
  - architecture, technologies, physical layer



- Challenges in IoT network & device management
  - Scalability, interoperability, energy efficiency, topology control, Quality of Service (QoS), fault tolerance, and security
- **M4DN.IoT** (Management for Device and Network in IoT): proposed platform for IoT networks & devices management
  - **M4DN.IoT** → outlines a management structure in two scopes:
  - **Local management :** nearby configure & monitor : Same ntk
  - **Remote management:** remote configure & monitor: different ntk



Internet of Things (IOT)

IoT Tools and Platforms

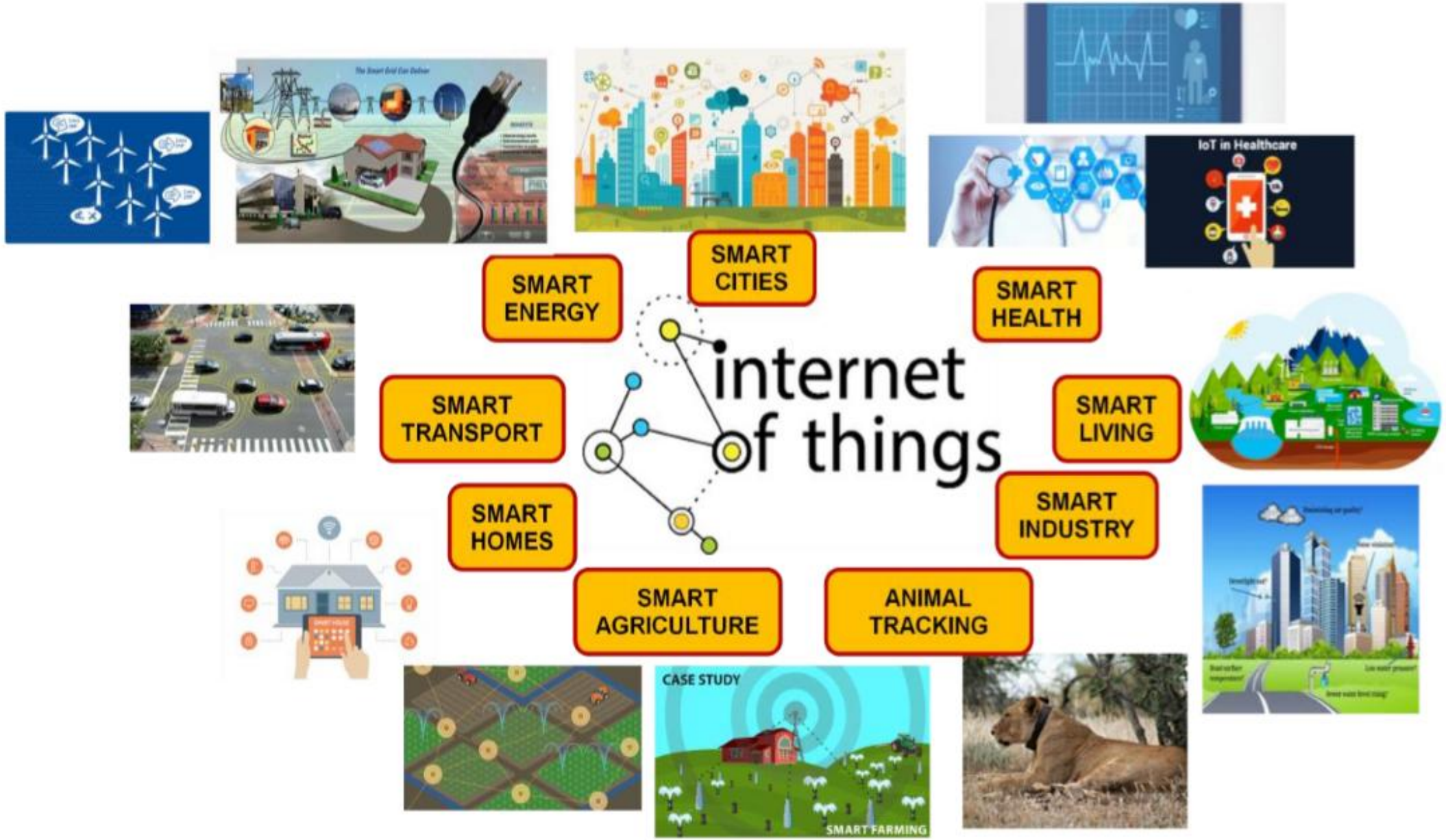
IoT Tools and Platforms

- IoT Platform are based on the Internet of Things and cloud technology

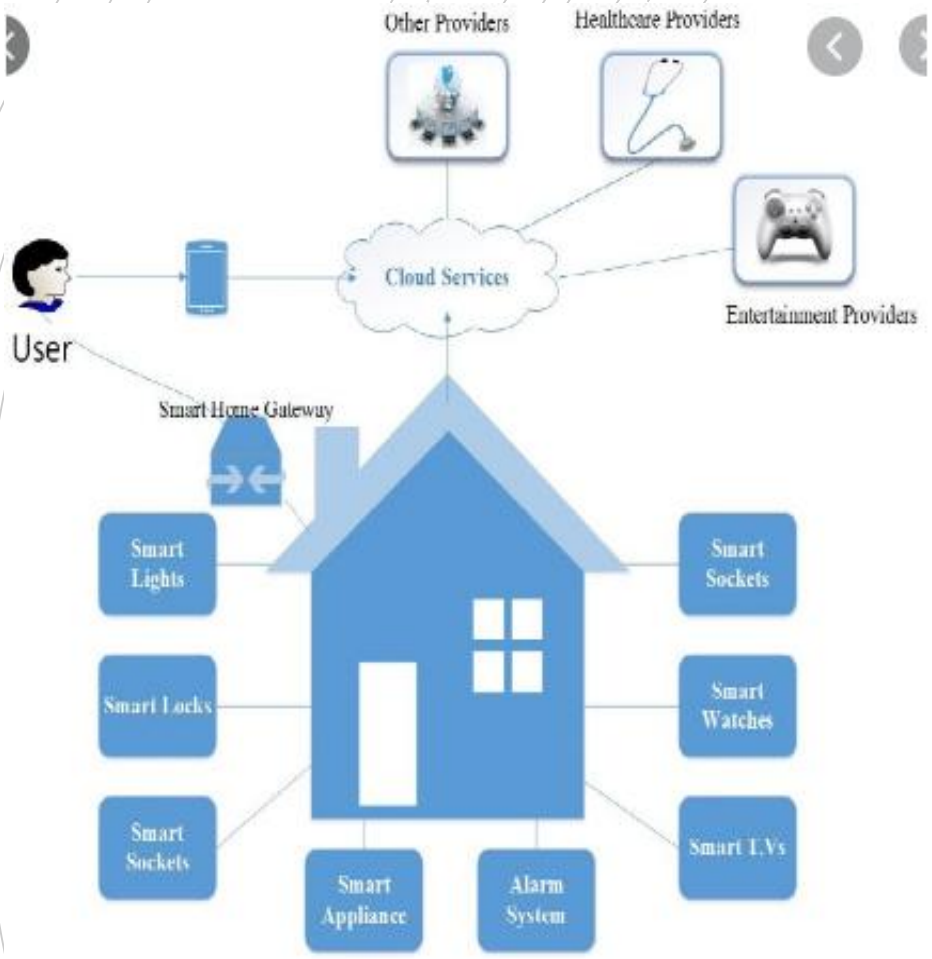
Some of the IOT platforms are

- **KAA** : Manage an unlimited number of connected devices
- **SiteWhere** : Add devices through self-registration
- **ThingSpeak**: Collect data in private channels and Share data with public channels
- **DeviceHive** : It supports Big data solutions like Apache Spark
- **Zetta** : allows you to assemble smartphone apps
- **ThingsBoard** : Real-time data visualization

IoT Across Application Domains

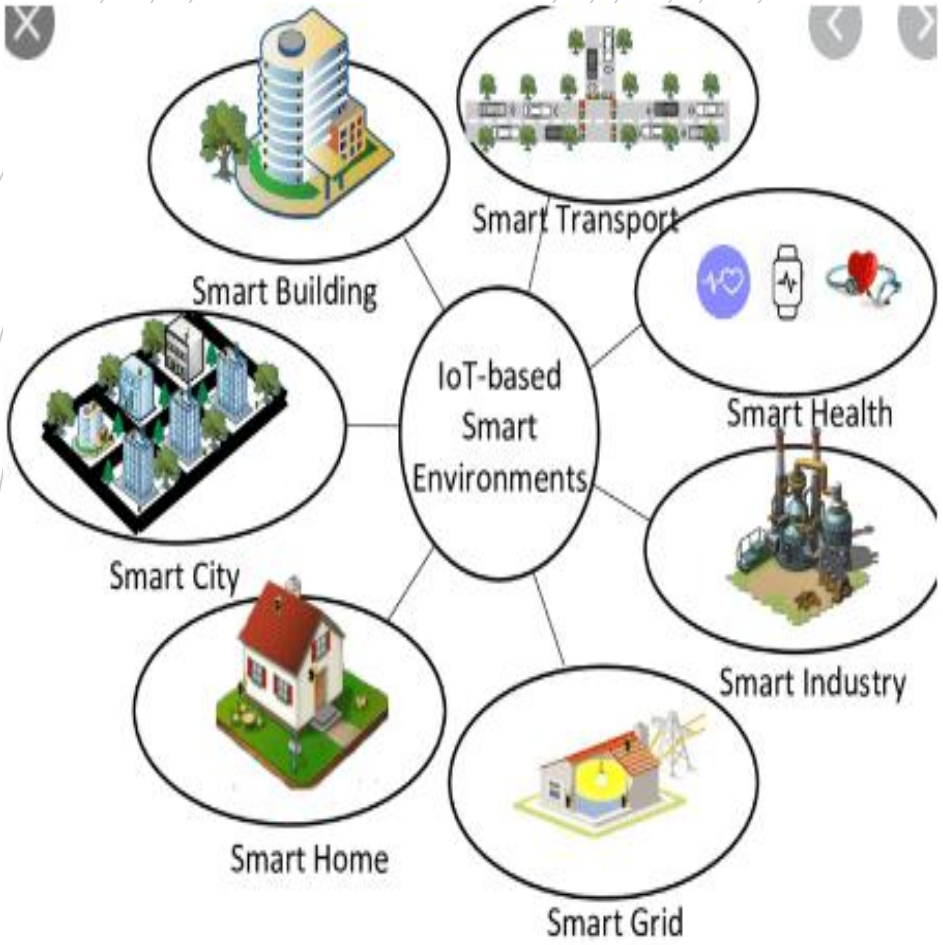


IoT Based Smart Home

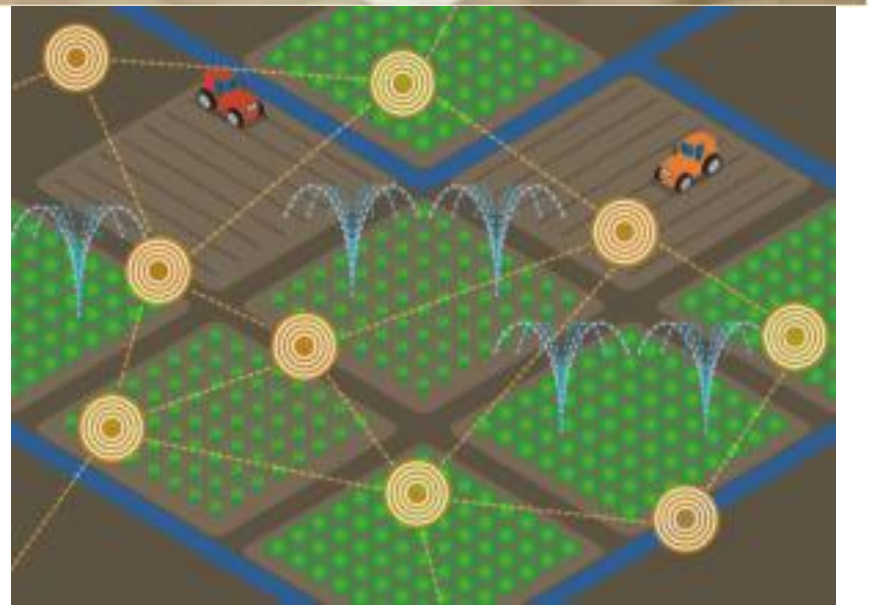


- It allows subscribers to remotely manage and monitor different home devices from anywhere via smartphones
- **Remote Control Appliances**: Switching on and off remotely appliances to avoid accidents and save energy.
- **Weather**: Displays outdoor weather conditions such as humidity, temperature, pressure, wind speed and rain levels with the ability to transmit data over long distance
- **Smart Home Appliances:**
- **Safety Monitoring**: cameras, and home alarm systems making people feel safe in their daily life at home.
- **Intrusion Detection Systems**: Detection of window and door openings and violations to prevent intruders.
- **Energy and Water Use**: Energy and water supply consumption monitoring to obtain advice on how to save cost and resources

IoT Based Smart City



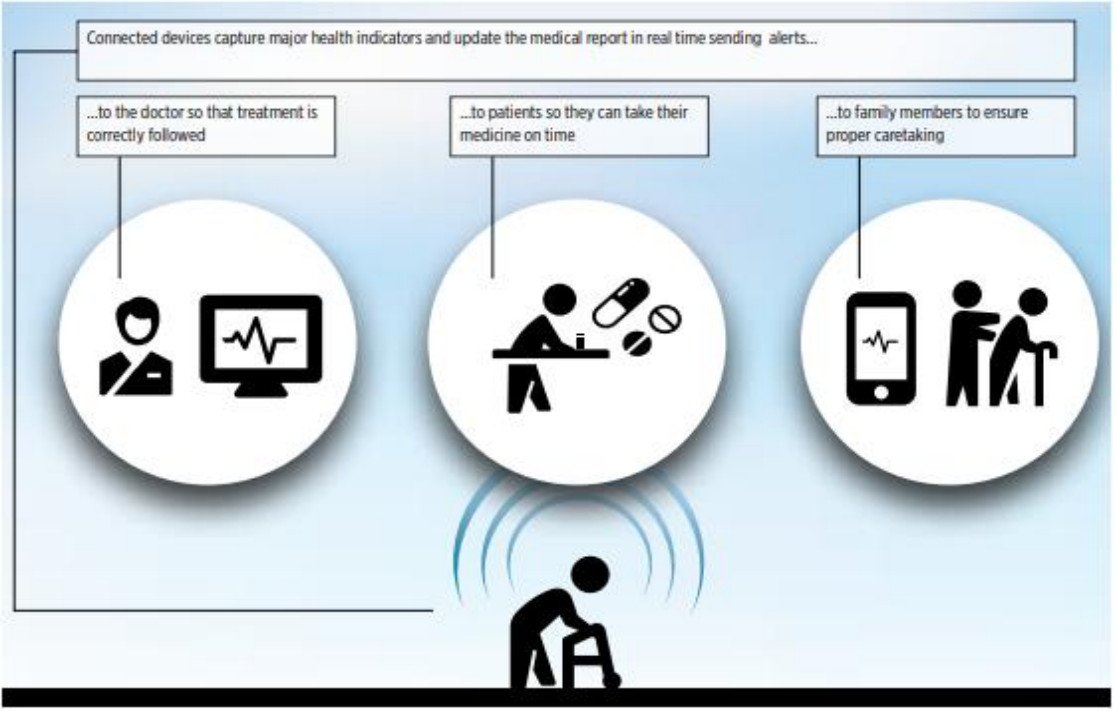
- **Structural Health**: Monitoring of vibrations and material conditions in buildings, bridges and historical monuments.
- **Lighting**: intelligent and weather adaptive lighting in street lights
- **Safety**: Digital video monitoring, fire control management, public announcement systems
- **Transportation**: Smart Roads and Intelligent High-ways with warning messages and diversions according to climate conditions and unexpected events like accidents or traffic jams.
- **Smart Parking**: Real-time monitoring of parking spaces available in the city making residents able to identify and reserve the closest available spaces
- **Waste Management**: Detection of rubbish levels in containers to optimize the trash collection routes

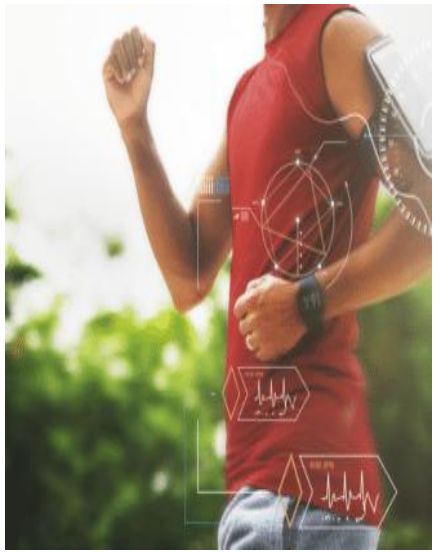


IoT Based Smart Farming

- A set of sensors such as light, humidity, temperature can be used to continuously monitor the field conditions
- **Green Houses**: Control micro-climate conditions to maximize the production of fruits and vegetables and its quality
- **Animal Farming/Tracking**: Location and identification of animals grazing in open pastures or location in big stables, Study of ventilation and air quality in farms and detection of harmful gases from excrements
- **Field Monitoring**: Reducing spoilage and crop waste with better monitoring, accurate ongoing data obtaining, and management of the agriculture fields, including better control of fertilizing, electricity and watering

IoT in Healthcare





- Wearable IoT devices let hospitals monitor their patients 'health at home, thereby reducing hospital stays while still providing up to the minute real-time information that could save lives.
- In hospitals, smart beds keep the staff informed as to the availability, thereby cutting wait time for free space.
- Putting IoT sensors on critical equipment means fewer breakdowns and increased reliability, which can mean the difference between life and death.
- Elderly care becomes significantly more comfortable with IoT.
- Real-time home monitoring, sensors can also determine if a patient has fallen or is suffering a heart attack

Additional Material…

Short YouTube Video Links on what IoT is…

- https://www.youtube.com/watch?v=6mBO2vqLv38
- https://www.youtube.com/watch?v=LlhmzVL5bm8

Home Work on IoT (5 pts.)

Detailed Description of a Selected IoT platform.

- Select one of the commonly known IoT platforms from this presentation or your own search.
- Provide a detailed description of the platform, including but not limited to:
  - general introduction,
  - the features included,
  - how it works,
  - possible application areas,
  - availability (opensource, subscription ….), …etc.
- Submission is in a handwritten copy (one page with no cover page).

Internet of Things (IOT)

Chapter Four

END